```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

importing all the titanic dataset

```
df=pd.read_csv("test.csv")
df
```

	PassengerId	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	892	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	NaN	Q
1	893	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	NaN	S
2	894	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	NaN	Q
3	895	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	NaN	S
4	896	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	NaN	S
...	...	...	...	...	...	...	...	...	...	...	...
413	1305	3	Spector, Mr. Woolf	male	NaN	0	0	A.5. 3236	8.0500	NaN	S
414	1306	1	Oliva y Ocana, Dona. Fermina	female	39.0	0	0	PC 17758	108.9000	C105	C
415	1307	3	Saether, Mr. Simon Sivertsen	male	38.5	0	0	SOTON/O.Q. 3101262	7.2500	NaN	S
416	1308	3	Ware, Mr. Frederick	male	NaN	0	0	359309	8.0500	NaN	S
417	1309	3	Peter, Master. Michael J	male	NaN	1	1	2668	22.3583	NaN	C

418 rows × 11 columns

```
df_train=pd.read_csv("train (1).csv")
df_train
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
...	...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7500	NaN	Q

891 rows × 12 columns

Merging both the dataset

```
df_merge=pd.concat([df,df_train])
df_merge
```

	PassengerId	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	Survived
0	892	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	NaN	Q	NaN
1	893	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	NaN	S	NaN
2	894	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	NaN	Q	NaN
3	895	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	NaN	S	NaN
4	896	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	NaN	S	NaN
...	...	...	...	...	...	...	...	...	...	...	...	...
886	887	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	NaN	S	0.0
887	888	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42	S	1.0
888	889	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	NaN	S	0.0
889	890	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C148	C	1.0
890	891	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7500	NaN	Q	0.0

1309 rows × 12 columns

If you want an overall description of the dataset or to have a basic understanding of the dataset we use { df.describe() }

```
df.describe()
```

	PassengerId	Pclass	Age	SibSp	Parch	Fare
count	418.000000	418.000000	332.000000	418.000000	418.000000	417.000000
mean	1100.500000	2.265550	30.272590	0.447368	0.392344	35.627188
std	120.810458	0.841838	14.181209	0.896760	0.981429	55.907576
min	892.000000	1.000000	0.170000	0.000000	0.000000	0.000000
25%	996.250000	1.000000	21.000000	0.000000	0.000000	7.895800
50%	1100.500000	3.000000	27.000000	0.000000	0.000000	14.454200
75%	1204.750000	3.000000	39.000000	1.000000	0.000000	31.500000
max	1309.000000	3.000000	76.000000	8.000000	9.000000	512.329200

we can also understand a single column using describe function by give the direct like to that column as df['x'], x = column name.

```
df['Age'].describe()
```

	Age
count	332.000000
mean	30.272590
std	14.181209
min	0.170000
25%	21.000000
50%	27.000000
75%	39.000000
max	76.000000

dtype: float64

UNIVARIATE ANALYSIS PROCESS ----Univariate analysis means analyzing one variable (column) at a time. The goal is to understand the distribution, central tendency, and spread of that single feature — without comparing it to others.:

It can be done on both Numerical columns and the categorical columns

UNIVARIATE ANALYSIS ON AGE COLUMN

```
df_merge['Age'].describe()
```

	Age
count	1046.000000
mean	29.881138
std	14.413493
min	0.170000
25%	21.000000
50%	28.000000
75%	39.000000
max	80.000000

dtype: float64

```
null_no=df_merge['Age'].isnull().sum()
print(null_no)
```

```
263
```

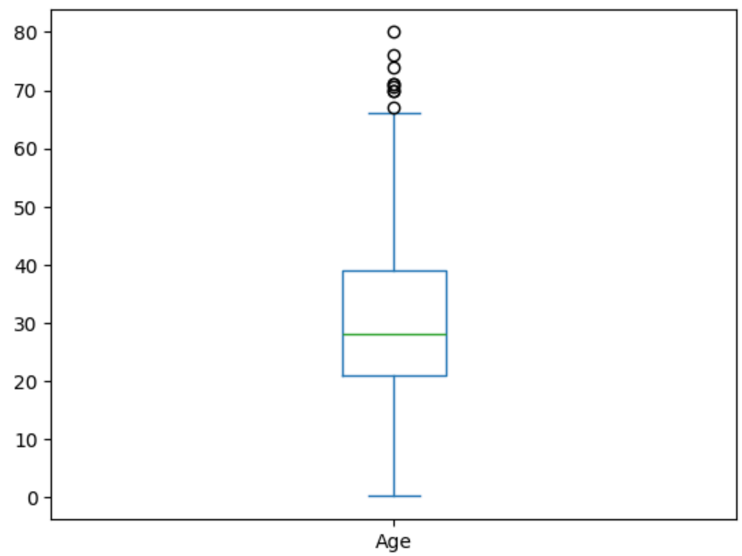
```
df_merge[df_merge['Age'].isnull()]
```

	PassengerId	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	Survived
10	902	3	Ilieff, Mr. Ylio	male	NaN	0	0	349220	7.8958	NaN	S	NaN
22	914	1	Flegenheim, Mrs. Alfred (Antoinette)	female	NaN	0	0	PC 17598	31.6833	NaN	S	NaN
29	921	3	Samaan, Mr. Elias	male	NaN	2	0	2662	21.6792	NaN	C	NaN
33	925	3	Johnston, Mrs. Andrew G (Elizabeth Lily" Watson)"	female	NaN	1	2	W./C. 6607	23.4500	NaN	S	NaN
36	928	3	Roth, Miss. Sarah A	female	NaN	0	0	342712	8.0500	NaN	S	NaN
...	...	...	...	...	...	...	...	...	...	...	...	...
859	860	3	Razi, Mr. Raihed	male	NaN	0	0	2629	7.2292	NaN	C	0.0
863	864	3	Sage, Miss. Dorothy Edith "Dolly"	female	NaN	8	2	CA. 2343	69.5500	NaN	S	0.0
868	869	3	van Melkebeke, Mr. Philemon	male	NaN	0	0	345777	9.5000	NaN	S	0.0
878	879	3	Laleff, Mr. Kristo	male	NaN	0	0	349217	7.8958	NaN	S	0.0
888	889	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	NaN	S	0.0

263 rows × 12 columns

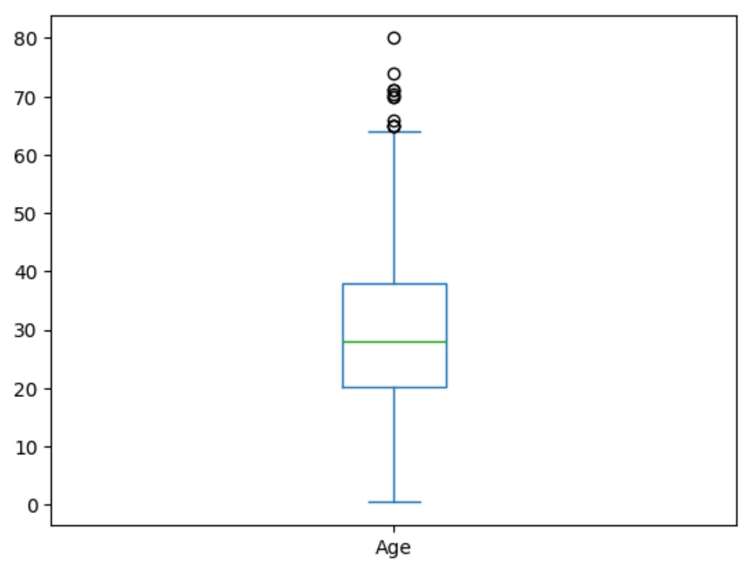
```
df_merge['Age'].plot(kind='box')
```

```
<Axes: >
```




```
df_train['Age'].plot(kind='box')
```

```
<Axes: >
```



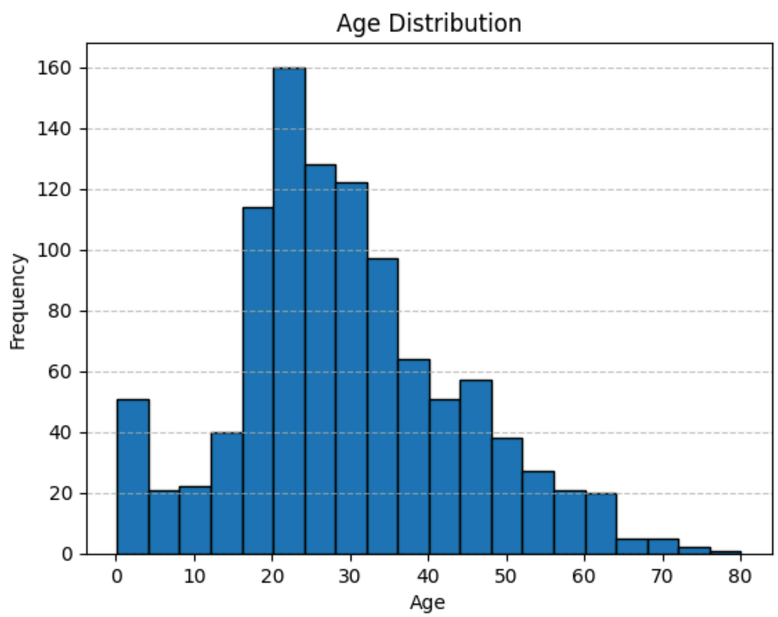

```
df_merge['Age'].max()
```

```
80.0
```

```
df_merge[df_merge['Age']>65]
```

	PassengerId	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	Survived
81	973	1	Straus, Mr. Isidor	male	67.0	1	0	PC 17483	221.7792	C55 C57	S	NaN
96	988	1	Cavendish, Mrs. Tyrell William (Julia Florence...	female	76.0	1	0	19877	78.8500	C46	S	NaN
33	34	2	Wheadon, Mr. Edward H	male	66.0	0	0	C.A. 24579	10.5000	NaN	S	0.0
96	97	1	Goldschmidt, Mr. George B	male	71.0	0	0	PC 17754	34.6542	A5	C	0.0
116	117	3	Connors, Mr. Patrick	male	70.5	0	0	370369	7.7500	NaN	Q	0.0
493	494	1	Artagaveytia, Mr. Ramon	male	71.0	0	0	PC 17609	49.5042	NaN	C	0.0
630	631	1	Barkworth, Mr. Algernon Henry Wilson	male	80.0	0	0	27042	30.0000	A23	S	1.0
672	673	2	Mitchell, Mr. Henry Michael	male	70.0	0	0	C.A. 24580	10.5000	NaN	S	0.0
745	746	1	Crosby, Capt. Edward Gifford	male	70.0	1	1	WE/P 5735	71.0000	B22	S	0.0
851	852	3	Svensson, Mr. Johan	male	74.0	0	0	347060	7.7750	NaN	S	0.0

```
df_merge['Age'].plot(kind='hist',bins=20,edgecolor='black')
plt.title('Age Distribution')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



```
counts,bin_edges=np.histogram(df_merge['Age'].dropna(),bins=20)
max_bin_index=np.argmax(counts)
lower_edge=bin_edges[max_bin_index]
upper_edge=bin_edges[max_bin_index+1]
print(f"Maximum number of passangers ({counts[max_bin_index]}) from {lower_edge:.1f} to {upper_edge:.1f}.")
```

```
Maximum number of passangers (160) from 20.1 to 24.1.
```

UNVARIATE ANALYSIS ON SURVIVED COLUMNS

Number of people Survived

```
No_of_Survived=(df_merge['Survived']>0).sum()
print(No_of_Survived)
```

```
342
```

Number of people Died

```
No_of_Died=(df_merge['Survived']<1).sum()
print(No_of_Died)
```

```
549
```

Missing Values

```
Null_value=df_merge['Survived'].isnull().sum()
print(Null_value)
```

```
418
```

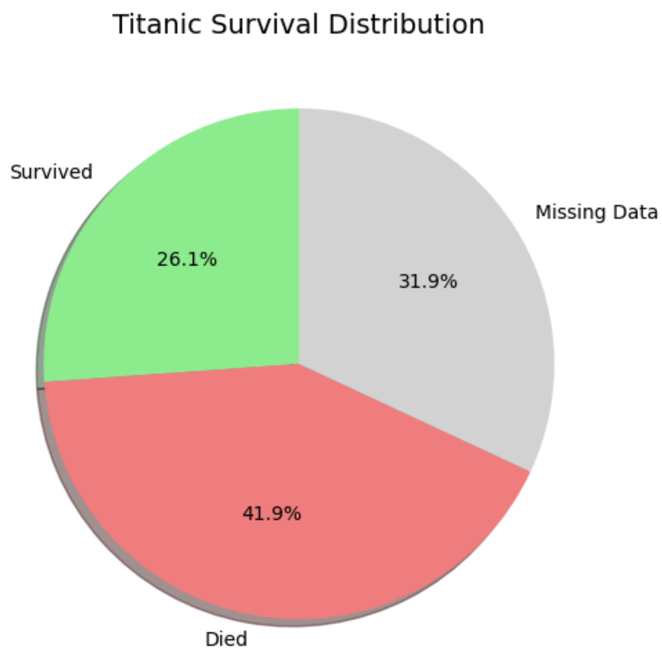
```
values = [No_of_Survived, No_of_Died, Null_value]
labels = ['Survived', 'Died', 'Missing Data']
```

```
plt.figure(figsize=(6,6))

plt.pie(
    values,
    labels=labels,
    autopct='%1.1f%%',
```

```
    startangle=90,
    colors=['lightgreen', 'lightcoral', 'lightgray'],
    shadow=True
)

plt.title('Titanic Survival Distribution', fontsize=14)
plt.show()
```




```
df_merge['Survived'].value_counts()
```

Survived	count
0.0	549
1.0	342

dtype: int64

Double-click (or enter) to edit

```
Survival_rate=df_merge['Survived'].mean()*100
print(f"Survival rate of passengers was {Survival_rate:.2f}%")
```

```
Survival rate of passengers was 38.38%
```

AS the survival rate was only 38.38% but its nots accurate as data of 418 passengers were missing.

UNIVARIATE ANALYSIS OF PCLASS

```
df_merge['Pclass'].value_counts(normalize=True)*100
```

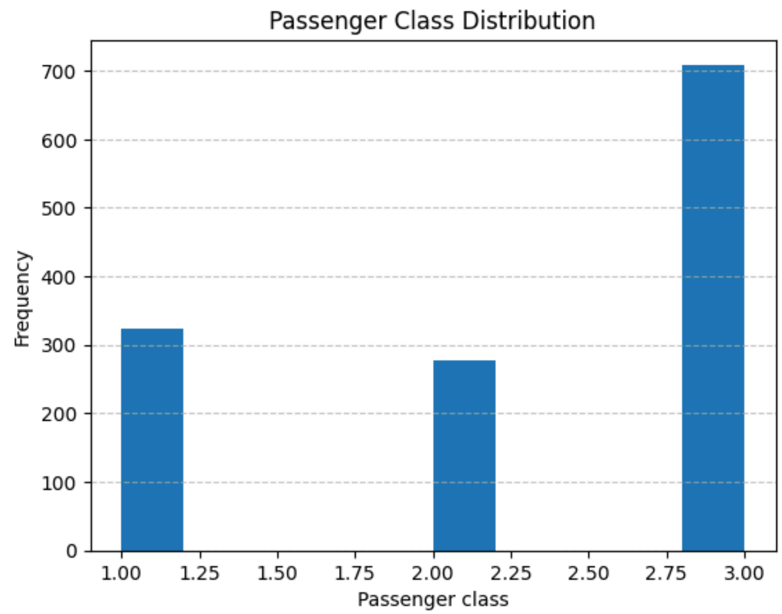
Pclass	proportion
3	54.163484
1	24.675325
2	21.161192

dtype: float64

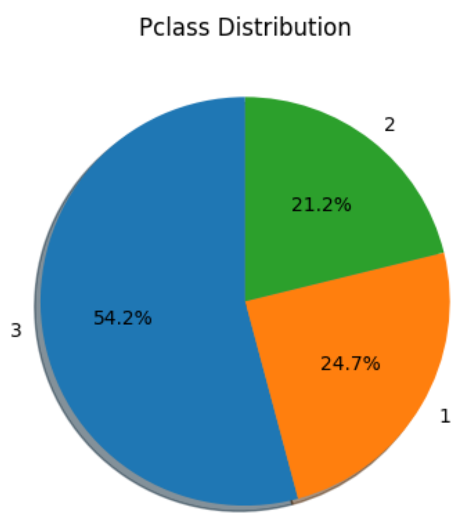
```
Missing_data=df_merge['Pclass'].isnull().sum()
print(Missing_data)
```

```
0
```

```
df_merge['Pclass'].plot(kind='hist')
plt.xlabel("Passenger class")
plt.title("Passenger Class Distribution")
plt.grid(axis='y', linestyle='--', alpha=0.7)
```

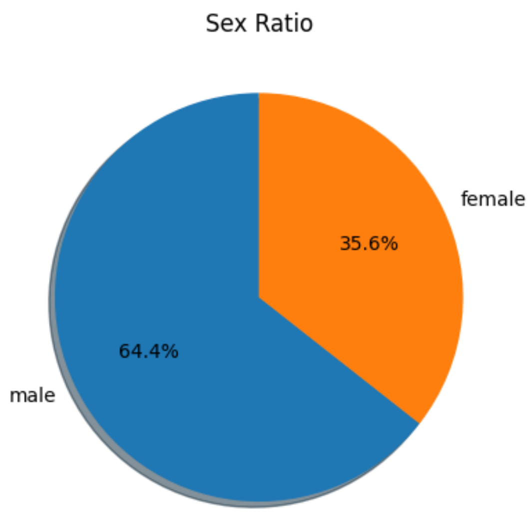



```
Pclass_values=df_merge['Pclass'].value_counts(normalize=True)*100
plt.pie(Pclass_values,labels=Pclass_values.index,autopct='%1.1f%%',startangle=90,shadow=True)
plt.title("Pclass Distribution")
plt.show()
```



UNIVARIATE ANALYSIS ON GENDER COLUMN

```
Table=df_merge['Sex'].value_counts(normalize=True)*100
plt.pie(Table,labels=Table.index,autopct='%1.1f%%',startangle=90,shadow=True)
plt.title('Sex Ratio')
plt.show()
```




```
is_null=df_merge['Sex'].isnull().sum()
print(is_null)
```

```
0
```

BIVARIATE ANALYSIS

Bivariate analysis is the statistical study of two variables to understand the relationship between them.

Lets do bivariate analysis on survived and other categorical column

```
results_1=pd.crosstab(df_merge['Survived'],df_merge['Pclass'],normalize='index')*100
print(results_1.round(2))
```

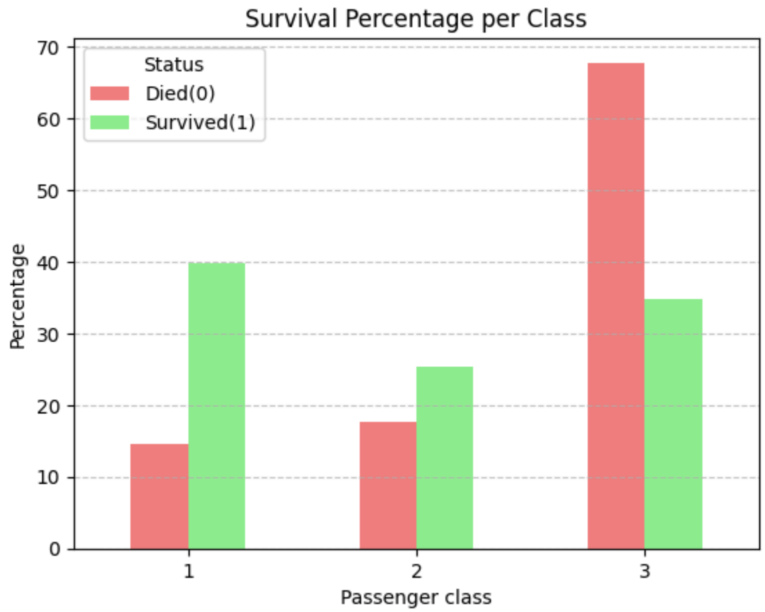
```
Pclass        1      2      3
Survived
0.0       14.57  17.67  67.76
1.0       39.77  25.44  34.80
```

here we can see that the total number of people died, 14.57% were from pclass 1 , 17.67% were from pclass 2 and maximum (67.76%) died from pclass 3.

```
results_1=pd.crosstab(df_merge['Survived'],df_merge['Pclass'],normalize='index')*100
results_1.T.plot(kind='bar',color=['lightcoral','lightgreen'])
```

```
plt.title("Survival Percentage per Class")
plt.xlabel("Passenger class")
plt.ylabel("Percentage")
plt.legend(title='Status',labels=['Died(0)','Survived(1)'])
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.xticks(rotation=0)
```

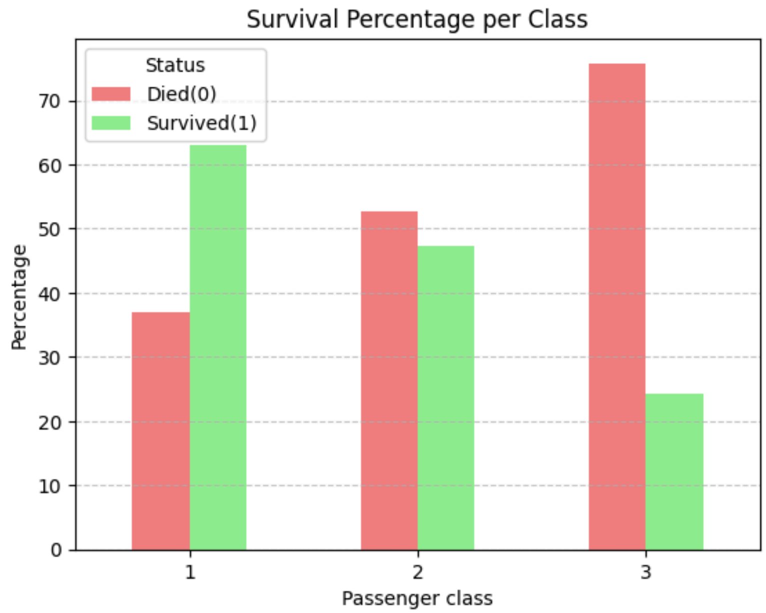
```
(array([0, 1, 2]), [Text(0, 0, '1'), Text(1, 0, '2'), Text(2, 0, '3')])
```



here we have distributed the total number of people survived and died based on the class.

```
results_2=pd.crosstab(df_merge['Survived'],df_merge['Pclass'],normalize='columns')*100
print(results_2.round(2))
results_2.T.plot(kind='bar',color=['lightcoral','lightgreen'])
plt.title("Survival Percentage per Class")
plt.xlabel("Passenger class")
plt.ylabel("Percentage")
plt.legend(title='Status',labels=['Died(0)','Survived(1)'])
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.xticks(rotation=0)
```

```
Pclass        1      2      3
Survived
0.0       37.04  52.72  75.76
1.0       62.96  47.28  24.24
(array([0, 1, 2]), [Text(0, 0, '1'), Text(1, 0, '2'), Text(2, 0, '3')])
```



And here we can see that from each Class what percent of people died and survivred

BIVARIATE ANALYSIS ON SURVIVED AND SEX COLUMN

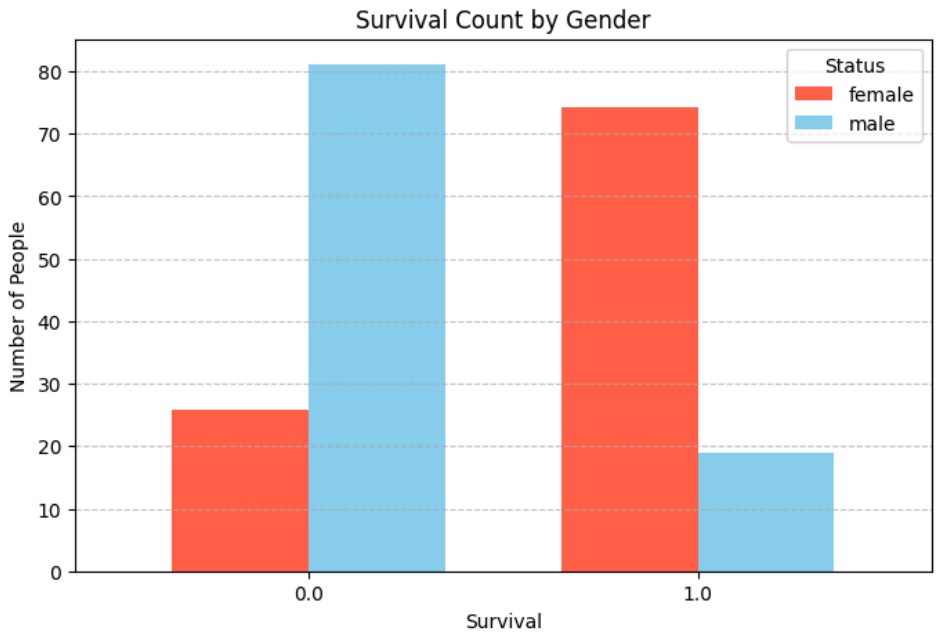
```
results_3=pd.crosstab(df_merge['Survived'],df_merge['Sex'],normalize='columns')*100
print(results_3)
```

```
Sex          female       male
Survived
0.0       25.796178  81.109185
1.0       74.203822  18.890815
```

```
ax = results_3.plot(
    kind='bar',
    figsize=(8,5),
    color=['tomato','skyblue'],
    width=0.7
)

plt.title("Survival Count by Gender")
plt.xlabel("Survival")
plt.ylabel("Number of People")
plt.xticks(rotation=0)
```

```
plt.legend(title="Status")
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



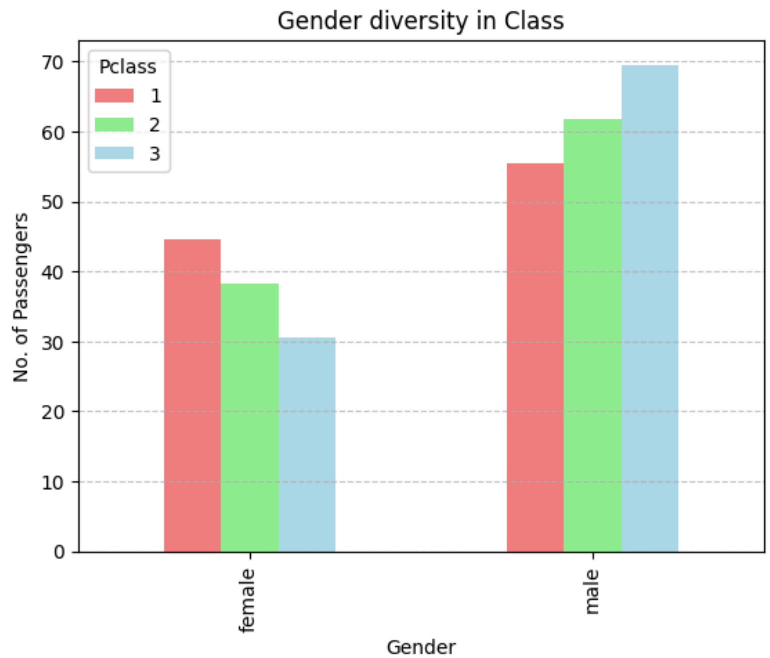
Here we can witness that majority of passenger were female who survived.

BIVARIATE ANALYSIS ON PCLASS AND SEX

```
results_4=pd.crosstab(df_merge['Pclass'],df_merge['Sex'],normalize='index')*100
print(results_4)
```

```
Sex        female       male
Pclass
1       44.582043  55.417957
2       38.267148  61.732852
3       30.465444  69.534556
```

```
results_4.T.plot(kind='bar',color=['lightcoral','lightgreen','lightblue'])
plt.title("Gender diversity in Class")
plt.xlabel("Gender")
plt.ylabel("No. of Passengers")
plt.grid(axis='y',linestyle='--',alpha=0.7)
```



with this we found that more female travelled with Pclass(1) and maximum male passengers were travelling from Pclass(3).

Insight-01

By studying graphs like "Gender diversity in class", "Survival count by gender", "Survival percentage by class", we can understand that majority of the passengers that survived were from Pclass(1) and that majority of them were female and maximum death count were from Pclass(3) and they were male . So it clearly shows that Pclass(1) females, were giving priority .

now using multiple columns to under deeper meaning.

lets find out that “Were girl children more likely to survive than boy children?”

```
df_merge['Child']=df_merge['Age']<18
```

Double-click (or enter) to edit

```
children_df=df_merge[df_merge['Child']==True]
children_df
```

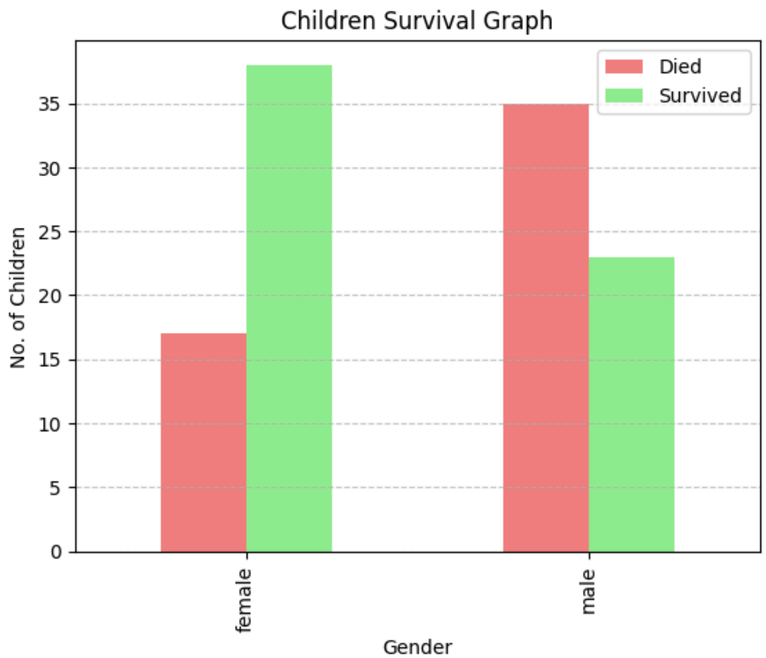
	PassengerId	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	Survived	Child
5	897	3	Svensson, Mr. Johan Cervin	male	14.0	0	0	7538	9.2250	NaN	S	NaN	True
21	913	3	Olsen, Master. Artur Karl	male	9.0	0	1	C 17368	3.1708	NaN	S	NaN	True
55	947	3	Rice, Master. Albert	male	10.0	4	1	382652	29.1250	NaN	Q	NaN	True
60	952	3	Dika, Mr. Mirko	male	17.0	0	0	349232	7.8958	NaN	S	NaN	True
64	956	1	Ryerson, Master. John Borie	male	13.0	2	2	PC 17608	262.3750	B57 B59 B63 B66	C	NaN	True
...	...	...	...	...	...	...	...	...	...	...	...	...	...
850	851	3	Andersson, Master. Sigvard Harald Elias	male	4.0	4	2	347082	31.2750	NaN	S	0.0	True
852	853	3	Boulos, Miss. Nourelain	female	9.0	1	1	2678	15.2458	NaN	C	0.0	True
853	854	1	Lines, Miss. Mary Conover	female	16.0	0	1	PC 17592	39.4000	D28	S	1.0	True
869	870	3	Johnson, Master. Harold Theodor	male	4.0	1	1	347742	11.1333	NaN	S	1.0	True
875	876	3	Najib, Miss. Adele Kiamie "Jane"	female	15.0	0	0	2667	7.2250	NaN	C	1.0	True

154 rows × 13 columns

```
results_5=pd.crosstab(children_df['Sex'],children_df['Survived'])
results_5.columns=['Died','Survived']
results_5
```

Sex	Died	Survived
female	17	38
male	35	23

```
results_5.plot(kind='bar',color=['lightcoral','lightgreen'])
plt.title("Children Survival Graph")
plt.xlabel("Gender")
plt.ylabel("No. of Children")
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



Insight-02

So here we can see that girl child had better chance of survival that boy child.

Bivariate analysis Embarked and Pclass Columns

```
df_merge['Embarked'].value_counts()
```

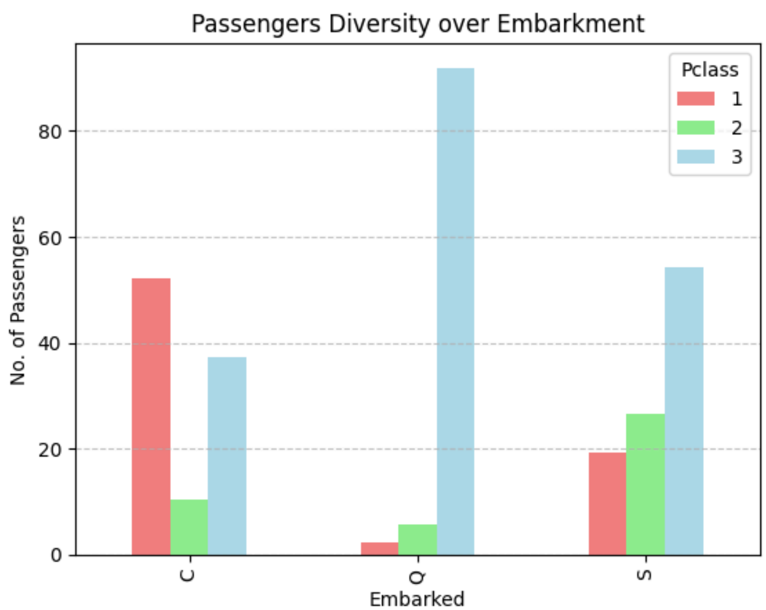
Embarked	count
S	914
C	270
Q	123

dtype: int64

```
results_6=pd.crosstab(df_merge['Embarked'],df_merge['Pclass'],normalize='index')*100
results_6
```

Pclass	1	2	3
Embarked			
C	52.222222	10.370370	37.407407
Q	2.439024	5.691057	91.869919
S	19.365427	26.477024	54.157549

```
results_6.plot(kind='bar',color=['Lightcoral','lightgreen','lightblue'])
plt.title("Passengers Diversity over Embarkment")
plt.ylabel("No. of Passengers")
plt.grid(axis='y',linestyle='--',alpha=0.7)
plt.show()
```



Here its clearly visible that majority of Pclass(1) Passengers were from southampton and majority of Pclass(3) Passengers were from Queenstown.

```
results_7=pd.crosstab(df_merge['Embarked'],df_merge['Sex'])
results_7
```

Sex	female	male
Embarked		
C	113	157
Q	60	63
S	291	623

Feature Engineering

Creating, modifying, or combining existing columns in your dataset to make new, more meaningful features that help your model or analysis.

```
df_merge['Family_size']=df_merge['Parch']+df_merge['SibSp']+1
```

```
df_merge['Family_size'].value_counts()
```

	count
Family_size	
1	790
2	235
3	159
4	43
6	25
5	22
7	16
11	11
8	8

dtype: int64

```
results_8=pd.crosstab(df_merge['Survived'],df_merge['Family_size'],normalize='columns')*100
results_8
```

Family_size	1	2	3	4	5	6	7	8	11
Survived									
0.0	69.646182	44.720497	42.156863	27.586207	80.0	86.363636	66.666667	100.0	100.0
1.0	30.353818	55.279503	57.843137	72.413793	20.0	13.636364	33.333333	0.0	0.0

```
df_merge[df_merge['Family_size']==11]
```

	PassengerId	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	Survived	Child	Family_size
188	1080	3	Sage, Miss. Ada	female	NaN	8	2	CA. 2343	69.55	NaN	S	NaN	False	11
342	1234	3	Sage, Mr. John George	male	NaN	1	9	CA. 2343	69.55	NaN	S	NaN	False	11
360	1252	3	Sage, Master. William Henry	male	14.5	8	2	CA. 2343	69.55	NaN	S	NaN	True	11
365	1257	3	Sage, Mrs. John (Annie Bullen)	female	NaN	1	9	CA. 2343	69.55	NaN	S	NaN	False	11
159	160	3	Sage, Master. Thomas Henry	male	NaN	8	2	CA. 2343	69.55	NaN	S	0.0	False	11
180	181	3	Sage, Miss. Constance Gladys	female	NaN	8	2	CA. 2343	69.55	NaN	S	0.0	False	11
201	202	3	Sage, Mr. Frederick	male	NaN	8	2	CA. 2343	69.55	NaN	S	0.0	False	11
324	325	3	Sage, Mr. George John Jr	male	NaN	8	2	CA. 2343	69.55	NaN	S	0.0	False	11
792	793	3	Sage, Miss. Stella Anna	female	NaN	8	2	CA. 2343	69.55	NaN	S	0.0	False	11
846	847	3	Sage, Mr. Douglas Bullen	male	NaN	8	2	CA. 2343	69.55	NaN	S	0.0	False	11
863	864	3	Sage, Miss. Dorothy Edith "Dolly"	female	NaN	8	2	CA. 2343	69.55	NaN	S	0.0	False	11

Insights-03

So chances of survival was less as family size increases as the both family with larger size died that night.

Multivariate Analysis

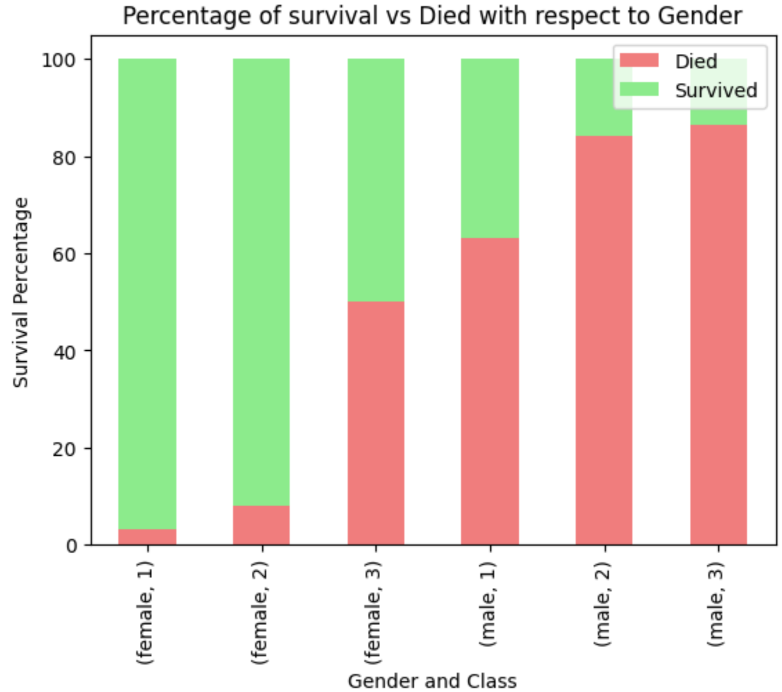
Analysis of pclass, Sex, Survived altogether.

```
results_9=pd.crosstab([df_merge['Sex'],df_merge['Pclass']],df_merge['Survived'],normalize='index')*100
results_9.columns=['Died','Survived']
results_9
```

Sex	Pclass	Died	Survived
female	**1**	3.191489	96.808511
	2	7.894737	92.105263
	3	50.000000	50.000000
male	**1**	63.114754	36.885246
	2	84.259259	15.740741
	3	86.455331	13.544669

```
results_9.plot(kind='bar',stacked='True',color=['lightcoral','lightgreen'])
plt.title("Percentage of survival vs Died with respect to Gender")
plt.xlabel("Gender and Class")
plt.ylabel("Survival Percentage")
```

```
Text(0, 0.5, 'Survival Percentage')
```



Let's see how did the family size by gender effected there chance of survival.

```
results_10=pd.crosstab([df_merge['Family_size'],df_merge['Sex']],df_merge['Survived'],normalize='index')*100
results_10.columns=['Died','Survived']
results_10
```

	Died	Survived